

CSE 4621: Machine Learning

Introduction to Machine

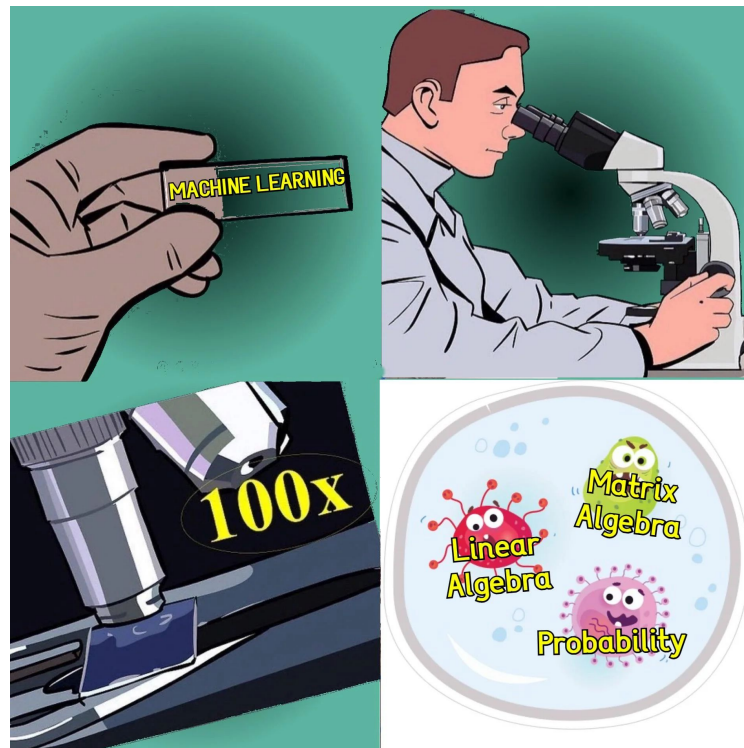
Learning

Lecture 1

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Lecture Outline

- What is machine learning?
 - Definitions and paradigms
 - History and milestones
 - Ethical and societal impact
- Mathematical Foundations
 - Linear algebra basics
 - Calculus and gradients
 - Probability and Statistics



What is Machine Learning?

ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



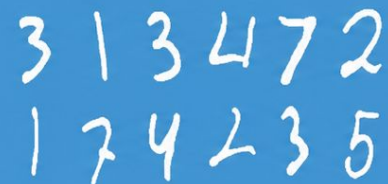
MACHINE LEARNING

Ability to learn without explicitly being programmed



DEEP LEARNING

Extract patterns from data using neural networks



What is Machine Learning?

- *Arthur Samuel (1959)*

“Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.”

- *Tom Mitchell (1998)*

“A computer program is said to **learn** from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

What is Machine Learning?

- A chess learning machine
 - T : Playing chess
 - P : Ratio of games won vs. total games
 - E : Playing chess against itself
- A self-driving machine
 - T : Driving a car using data from sensors
 - P : Average distance travelled before an error (judged by humans)
 - E : Observing a human driving and recoding their input

Raw Data

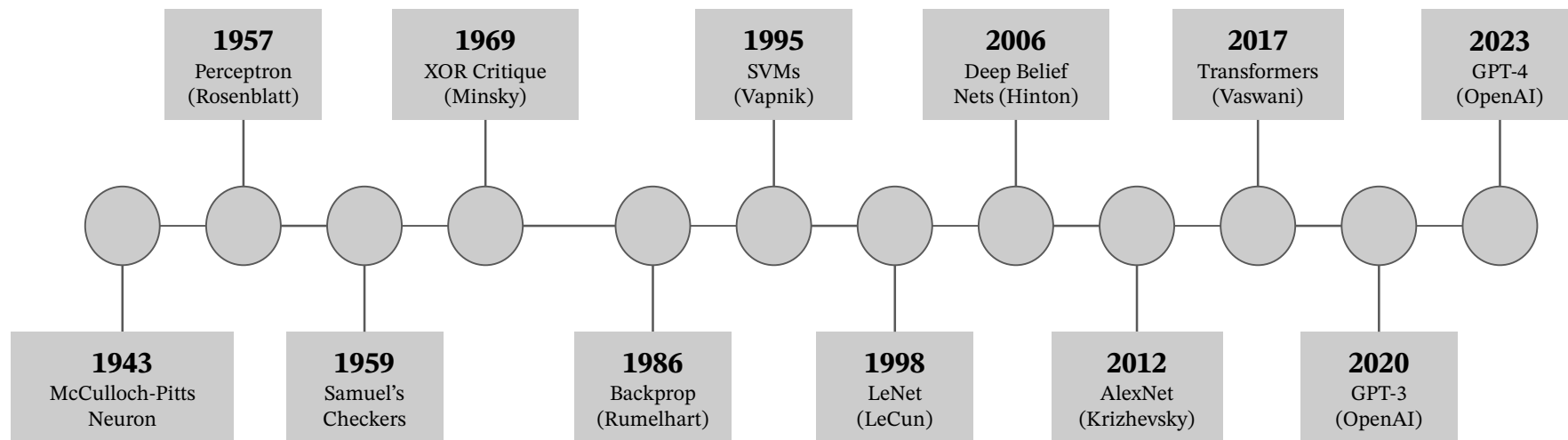
Representation

ML Model

Predictions

Feedback

A Brief History of Machine Learning



1943-1969: Foundations

Formal neuron models, first learning algorithms; naive optimism followed by the first “AI winter”.

1970-2005: Revival

Backprop rescues neural networks; kernel methods dominate; Statistical ML matures.

2006-Present: Deep Learning

GPU compute + Big data + Investment

Ethical and Societal Impact

- Bias and Fairness
 - Models trained on biased data perpetuate discrimination (facial recognition, hiring algorithms).
- Privacy
 - Large models memorise training data; inference attacks can recover sensitive records.
- Interpretability
 - “Black-box” decisions in medicine or law raise accountability concerns.
- Environmental Cost
 - Training GPT-4 class models $\approx$ 500 tonnes CO₂ equivalent.
 - If ChatGPT handles 1 billion queries daily, total water consumption can reach around 320,000 liters to over 100 million liters per day.

Paradigms of Machine Learning

Supervised Learning

Input: Labelled Pairs (x_i, y_i)

Goal: Learn $f: x \rightarrow y$

Examples:

- Classification
- Regression

Algorithms:

- Linear Regression
- SVMs
- Neural Networks

Unsupervised Learning

Input: Unlabelled Data (x_i)

Goal: Find hidden pattern

Examples:

- Clustering
- Dimension Reduction

Algorithms:

- K-means
- PCA
- Autoencoders

Reinforcement Learning

Input: Reward/Punishment for actions

Goal: Maximize cumulative reward

Examples:

- Game Playing
- Robotic Traversal

Algorithms:

- Q-Learning
- Policy Gradient

Mathematical Foundations - Linear Algebra

A matrix encodes a linear transformation: $Ax = b$

Key operations in Machine Learning:

- Matrix multiplication $\rightarrow$ Feature transformation
- Dot product $\rightarrow$ Weighted sum
- Diagonalization $\rightarrow A = Q\Lambda Q^T$
- Singular Value Decomposition $\rightarrow A = U\Sigma V^T$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ \vdots \\ \vdots \\ x_d \end{bmatrix} \in \mathbb{R}^d$$

$$\|x\|_2 = \sqrt{\sum_{i=1}^d x_i^2}$$

Mathematical Foundations - Calculus

A derivative represents the **instantaneous rate of change** of a function with respect to one of its variables

- Derivative of a constant function is always zero
- Power rule
- Sum/Difference rule
- Product rule
- Chain rule
- Partial derivatives
- Local/Global maxima or minima

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

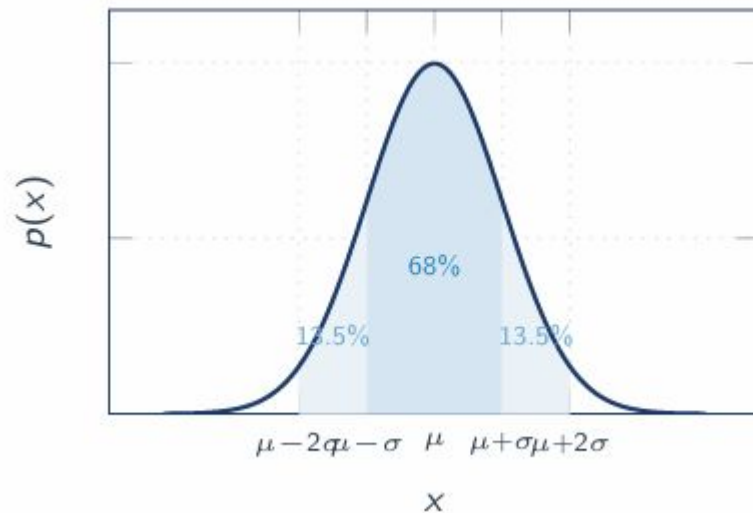
Mathematical Foundations - Probability and Statistics

- Probability: $P(A) \in [0, 1]$
- Joint probability: $P(X, Y) = P(X | Y)P(Y)$

- Bayes' theorem: $P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)}$

- Expectation and Variance

$$\mathbb{E}[X] = \int x p(x) dx, \quad \text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$



Textbook Chapters

- [Goodfellow] - Chapter 1, Chapter 2, Chapter 3
- [Bishop] Chapter 1, Chapter 2,
- [Alpaydin] Chapter 1

Recommended Resources

1. [Weapons of Math Destruction | Cathy O'Neil | Talks at Google](#)
2. [Essence of linear algebra | 3Blue1Brown](#)
3. [Essence of calculus | 3Blue1Brown](#)
4. [Statistics fundamentals | StatQuest](#)
5. [Linear Transformation Visualization](#)
6. [Probability Bootcamp | Steve Brunton](#)

Thank you